AI in Industry, ETH Zurich, autumn 2026

# The Word,



Rembrandt van Rijn, *A Scholar in His Study ('Faust')*, c. 1652. Etching, drypoint and burin. The Metropolitan Museum of Art, New York. CC0

AI in Industry, ETH Zurich, autumn 2026

# The Word, the Thought,



Rembrandt van Rijn, *A Scholar in His Study ('Faust')*, c. 1652. Etching, drypoint and burin. The Metropolitan Museum of Art, New York. CC0

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# The Word, the Thought, the Power,



Rembrandt van Rijn, *A Scholar in His Study ('Faust')*, c. 1652. Etching, drypoint and burin. The Metropolitan Museum of Art, New York. CC0

AI in Industry, ETH Zurich, autumn 2026

# The Word, the Thought, the Power, the Act

*Four blocks, four voices, one order*

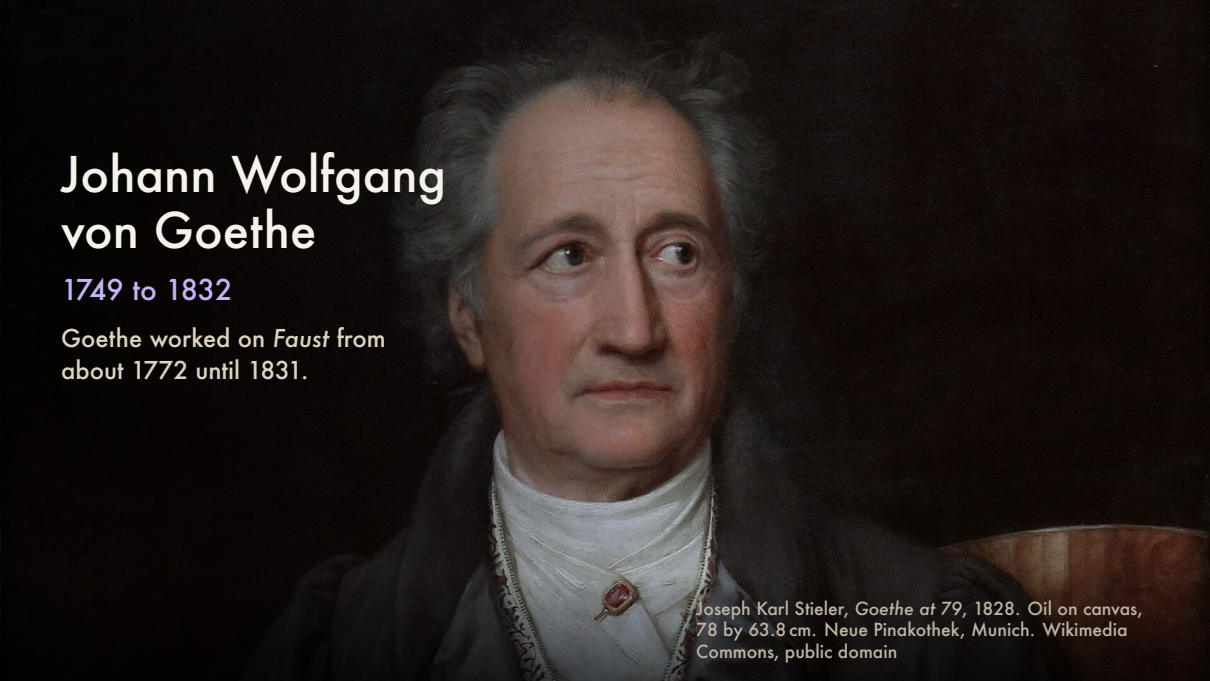


Rembrandt van Rijn, *A Scholar in His Study ('Faust')*, c. 1652. Etching, drypoint and burin. The Metropolitan Museum of Art, New York. CC0

# Johann Wolfgang von Goethe

1749 to 1832

Goethe worked on *Faust* from about 1772 until 1831.



Joseph Karl Stieler, *Goethe at 79*, 1828. Oil on canvas, 78 by 63.8 cm. Neue Pinakothek, Munich. Wikimedia Commons, public domain

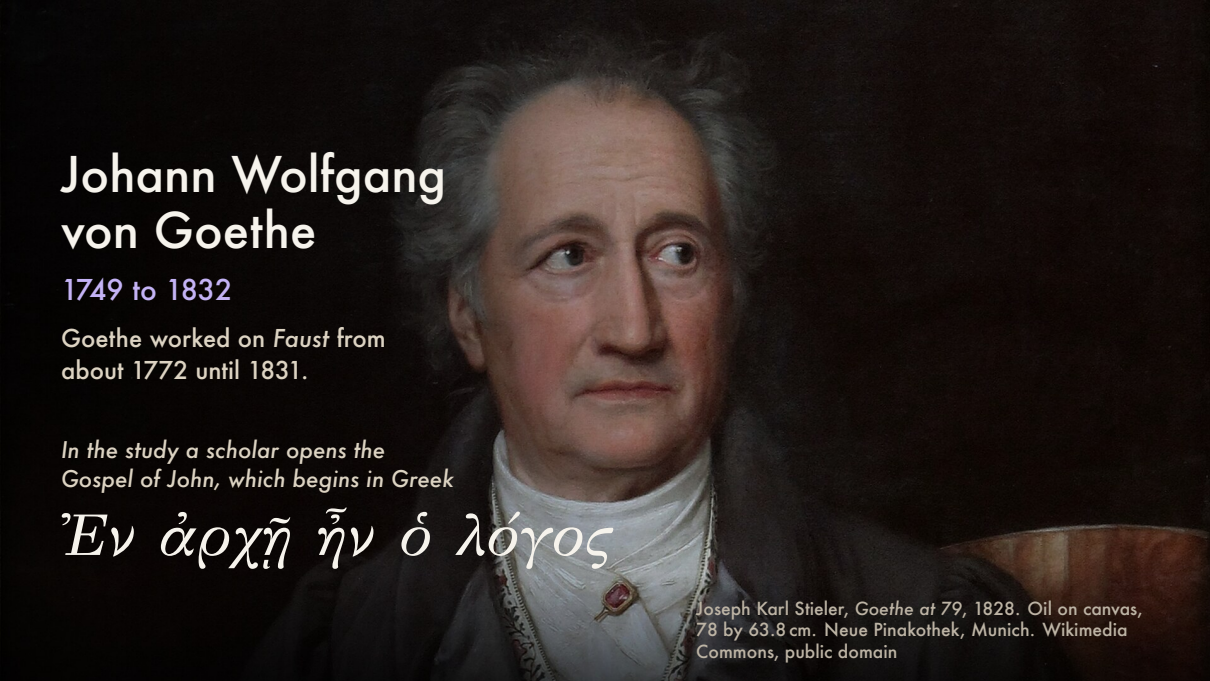
# Johann Wolfgang von Goethe

1749 to 1832

Goethe worked on *Faust* from about 1772 until 1831.

*In the study a scholar opens the Gospel of John, which begins in Greek*

Ἐν ἀρχῇ ἦν ὁ λόγος



Joseph Karl Stieler, *Goethe at 79*, 1828. Oil on canvas, 78 by 63.8 cm. Neue Pinakothek, Munich. Wikimedia Commons, public domain



Im Anfang war das

# Wort.

*In the Beginning*
*was the Word.*

Goethe, *Faust I*, the study. English by Bayard Taylor

Rembrandt van Rijn, *A Scholar in His Study ('Faust')*, c. 1652. The Metropolitan Museum of Art, New York. CC0



Im Anfang war das

# Wort.

*In the Beginning was the Word.*

Im Anfang war der

# Sinn.

*In the Beginning was the Thought.*

Goethe, *Faust I*, the study. English by Bayard Taylor

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Im Anfang war das

# Wort.

*In the Beginning was the Word.*

Im Anfang war der

# Sinn.

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Im Anfang war die

# Kraft.

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Im Anfang war das

# Wort.

*In the Beginning was the Word.*

Im Anfang war der

# Sinn.

*In the Beginning was the Thought.*

Im Anfang war die

# Kraft.

*In the Beginning was the Power.*

Im Anfang war die

# Tat.

*In the Beginning was the Act.*

Goethe, *Faust I*, the study. English by Bayard Taylor

Rembrandt van Rijn, *A Scholar in His Study ('Faust')*, c. 1652. The Metropolitan Museum of Art, New York. CC0

# Xunzi

c. 310 to after 238 BC

He warned that each step taken alone fails: *hearing* without *seeing*, *seeing* without *knowing*, *knowing* without *practice*.

Xunzi, chapter 8, 儒效. English by the lecturer

Xunzi, Qing dynasty portrait, from an album of portraits of sages. Wikimedia Commons, public domain



不聞不若聞之，聞之不若見之，
見之不若知之，知之不若行之，
學至於行之而止矣。

*Each is better than the one before. Learning ends in practice.*

Xunzi, chapter 8. English: lecturer

Xunzi, chapter 8, with the commentary of Yang Liang. Song printed edition. National Central Library, Taiwan. Public domain

不聞不若聞之，聞之不若見之，
見之不若知之，知之不若行之，
學至於行之而止矣。

*Each is better than the one before. Learning ends in practice.*

## 聞
## Hearing

Xunzi, chapter 8. English: lecturer

Xunzi, chapter 8, with the commentary of Yang Liang. Song printed edition. National Central Library, Taiwan. Public domain

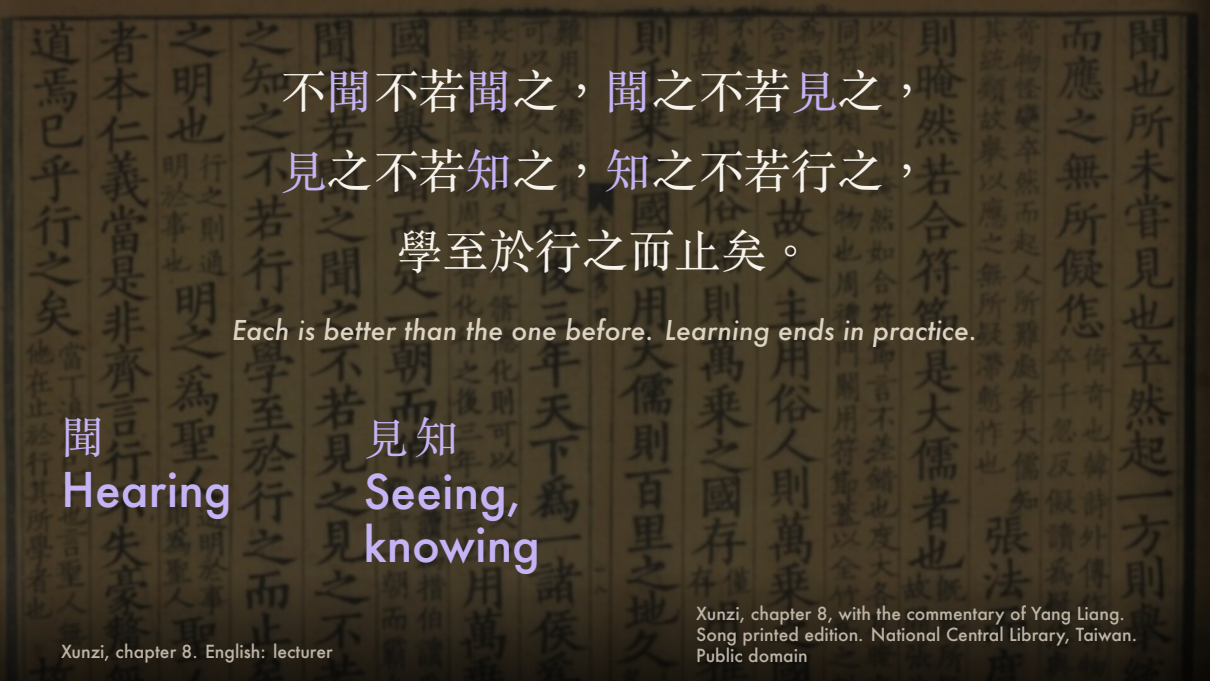
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*Each is better than the one before. Learning ends in practice.*

聞
Hearing

見 知
Seeing, knowing

Xunzi, chapter 8. English: lecturer



Xunzi, chapter 8, with the commentary of Yang Liang. Song printed edition. National Central Library, Taiwan. Public domain

不聞不若聞之，聞之不若見之，
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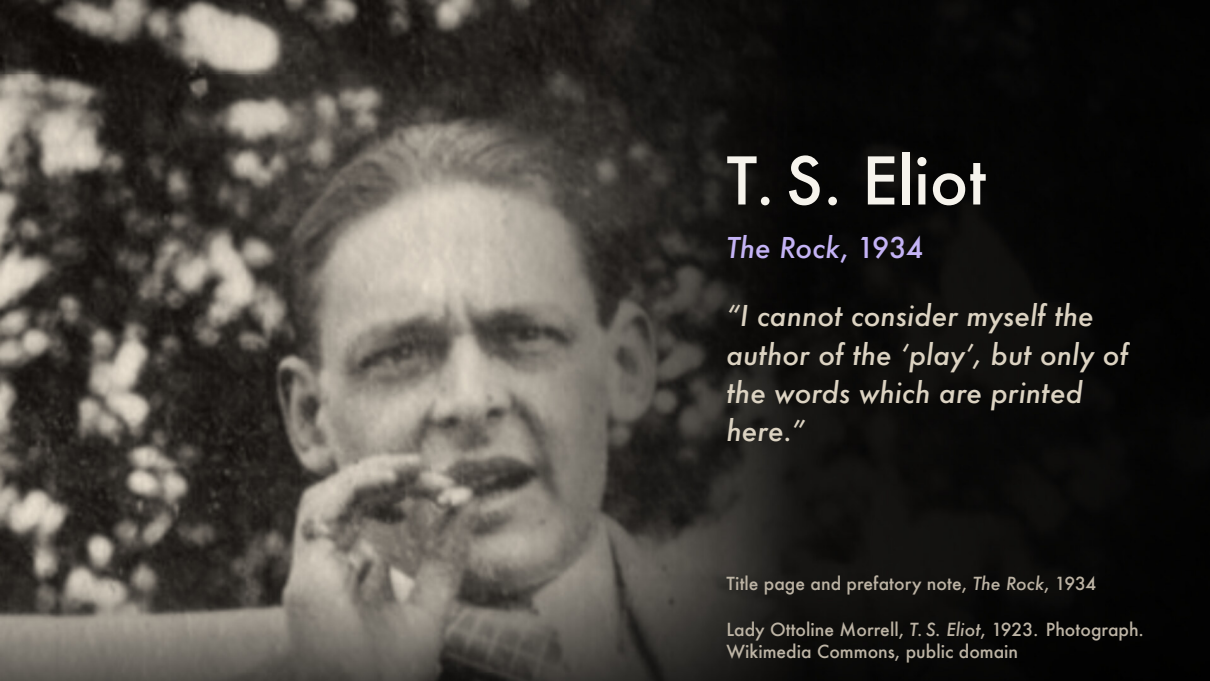
聞
Hearing

見 知
Seeing, knowing

行
Practice

Xunzi, chapter 8. English: lecturer

Xunzi, chapter 8, with the commentary of Yang Liang. Song printed edition. National Central Library, Taiwan. Public domain



# T. S. Eliot

*The Rock*, 1934

*"I cannot consider myself the author of the 'play', but only of the words which are printed here."*

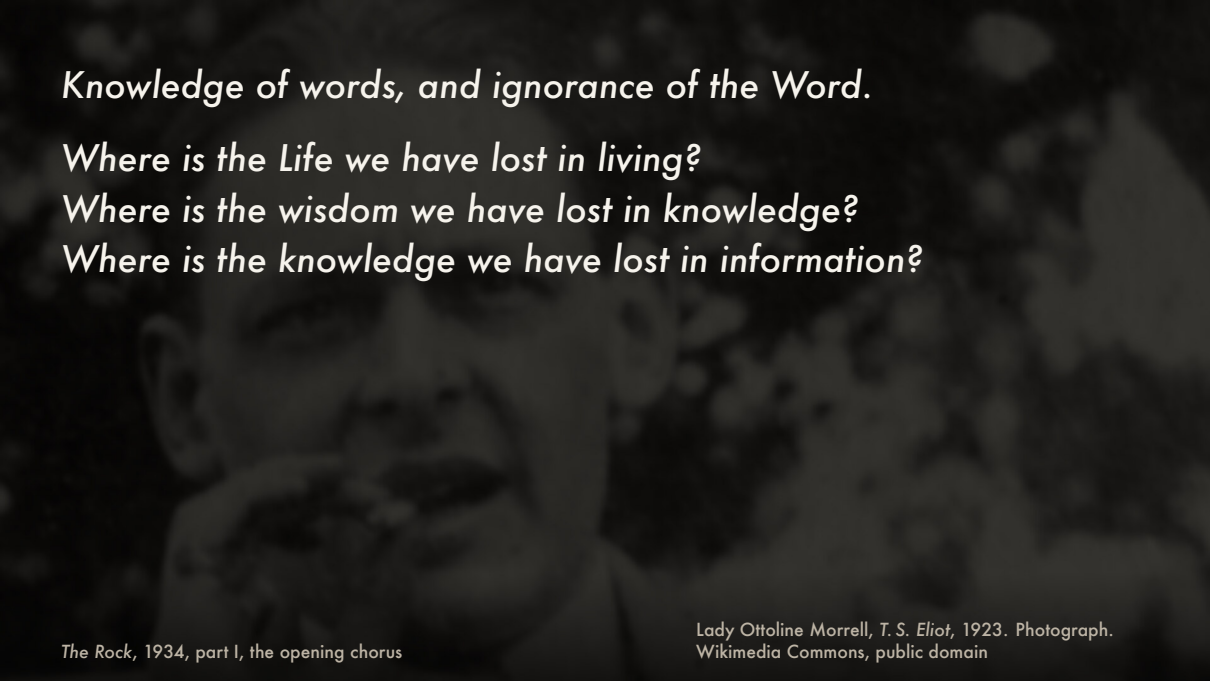
Title page and prefatory note, *The Rock*, 1934

Lady Ottoline Morrell, *T. S. Eliot*, 1923. Photograph. Wikimedia Commons, public domain

*Knowledge of words, and ignorance of the Word.*

*Where is the Life we have lost in living?*
*Where is the wisdom we have lost in knowledge?*
*Where is the knowledge we have lost in information?*

*The Rock*, 1934, part I, the opening chorus



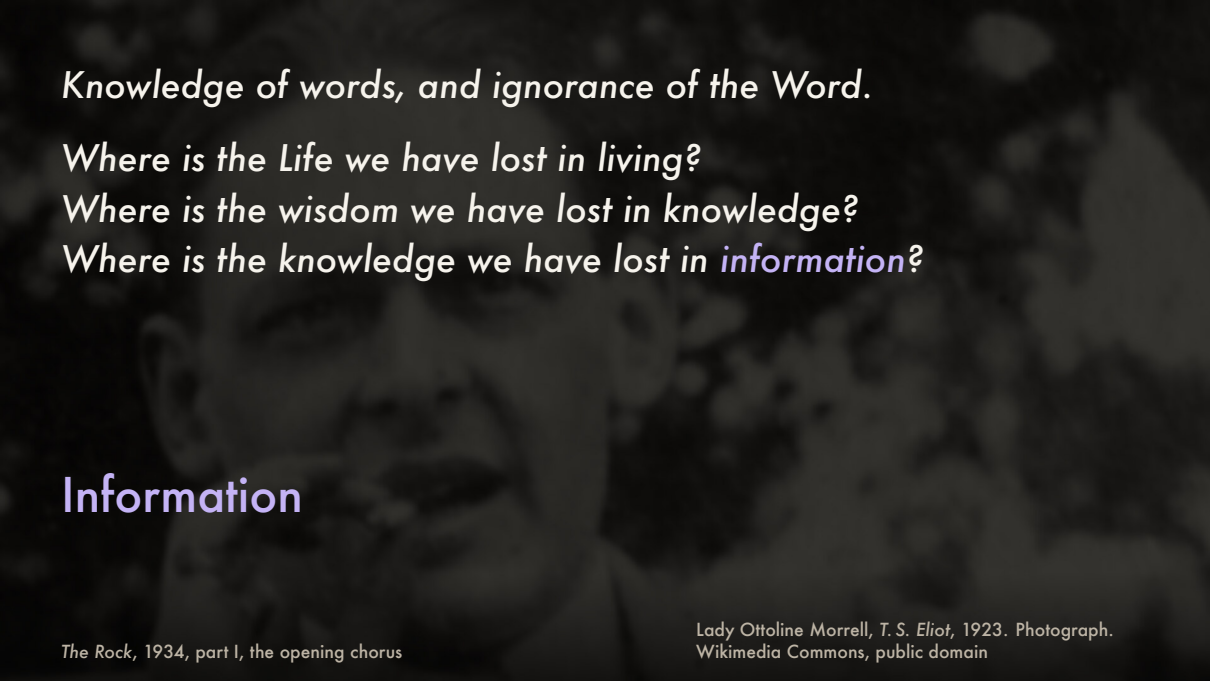
Lady Ottoline Morrell, *T. S. Eliot*, 1923. Photograph. Wikimedia Commons, public domain

*Knowledge of words, and ignorance of the Word.*

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# Information

*The Rock*, 1934, part I, the opening chorus



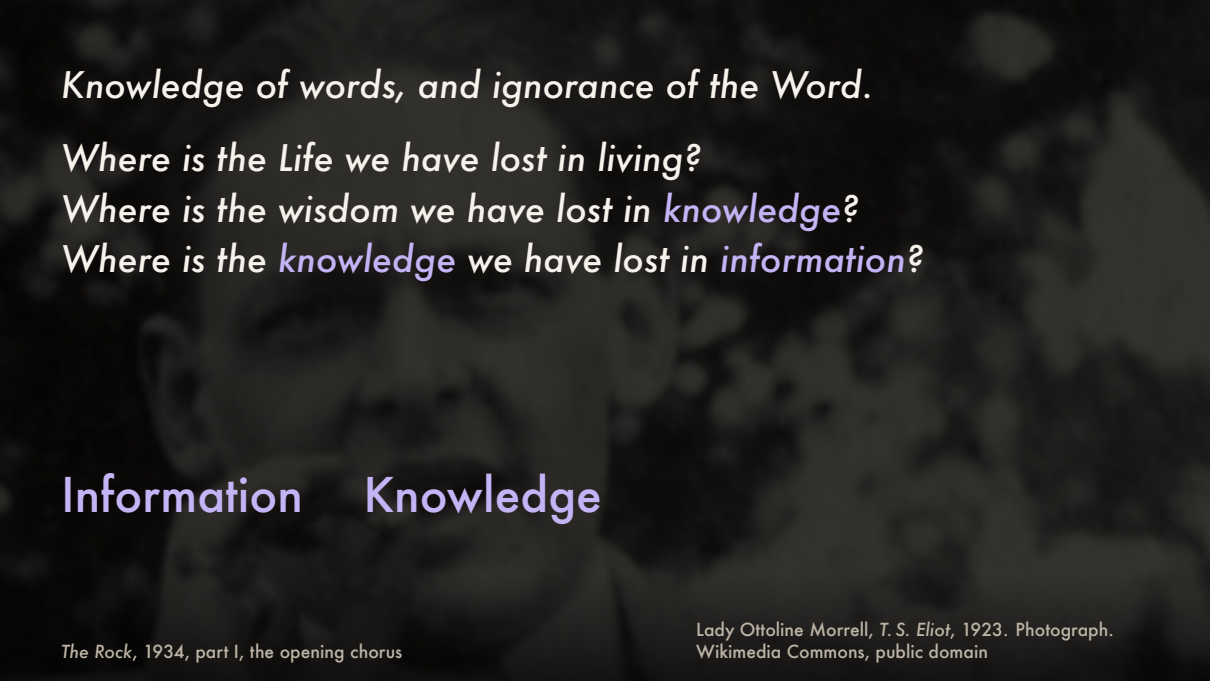
Lady Ottoline Morrell, *T. S. Eliot*, 1923. Photograph. Wikimedia Commons, public domain

*Knowledge of words, and ignorance of the Word.*

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*Where is the knowledge we have lost in information?*

Information Knowledge

*The Rock*, 1934, part I, the opening chorus



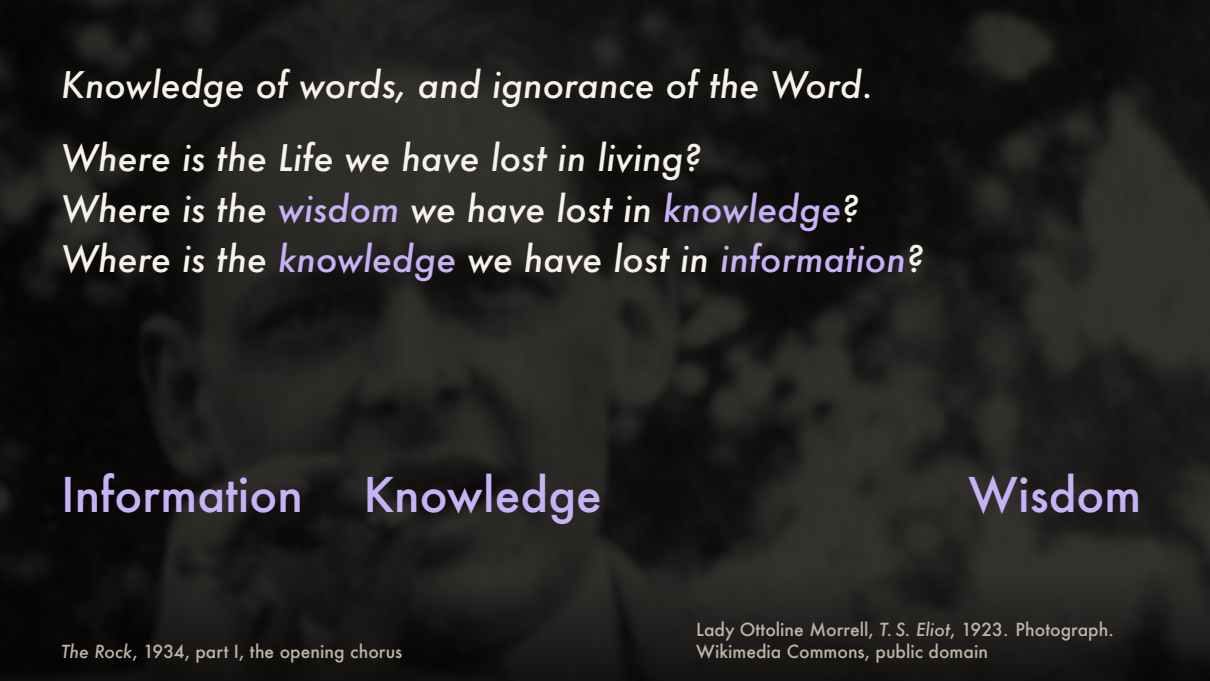
Lady Ottoline Morrell, *T. S. Eliot*, 1923. Photograph. Wikimedia Commons, public domain

*Knowledge of words, and ignorance of the Word.*

*Where is the Life we have lost in living?*
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*Where is the knowledge we have lost in information?*

Information Knowledge Wisdom

*The Rock*, 1934, part I, the opening chorus



Lady Ottoline Morrell, *T. S. Eliot*, 1923. Photograph. Wikimedia Commons, public domain



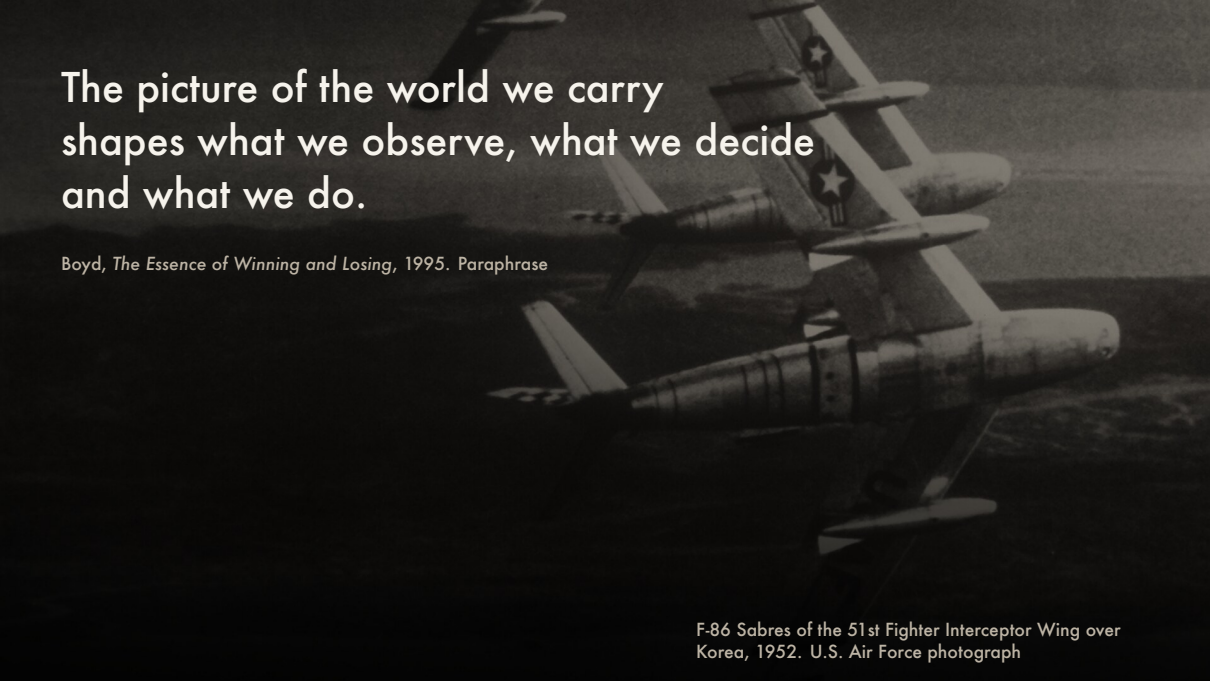
U.S. Air Force. *John Boyd, probably in Korea*, 1953. Photograph. U.S. Government. Public domain

# John Boyd

1927 to 1997

Boyd arrived in Korea in March 1953 and flew 22 missions in the F-86 Sabre, in which he did not fire his guns or score a kill. In 1960 he wrote the Air Force's manual of air combat tactics.

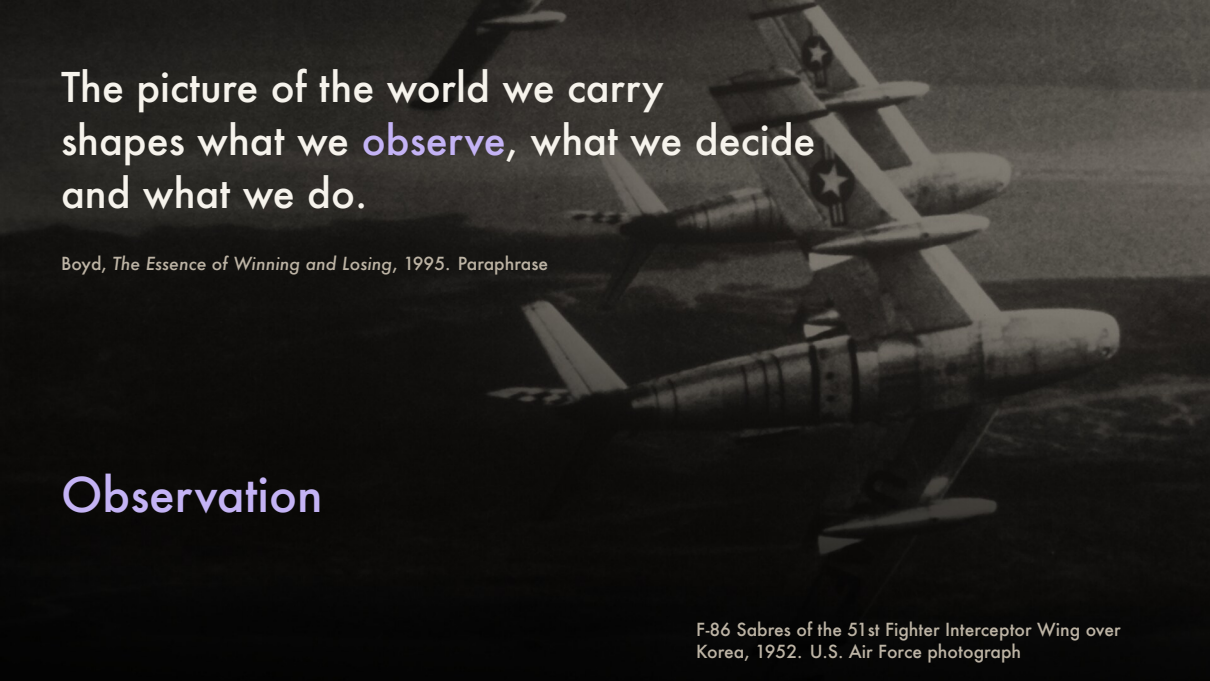
Wikipedia



The picture of the world we carry shapes what we observe, what we decide and what we do.

Boyd, *The Essence of Winning and Losing*, 1995. Paraphrase

F-86 Sabres of the 51st Fighter Interceptor Wing over Korea, 1952. U.S. Air Force photograph



The picture of the world we carry shapes what we observe, what we decide and what we do.

Boyd, *The Essence of Winning and Losing*, 1995. Paraphrase

# Observation

F-86 Sabres of the 51st Fighter Interceptor Wing over Korea, 1952. U.S. Air Force photograph



The picture of the world we carry shapes what we observe, what we decide and what we do.

Boyd, *The Essence of Winning and Losing*, 1995. Paraphrase

Observation Orientation

F-86 Sabres of the 51st Fighter Interceptor Wing over Korea, 1952. U.S. Air Force photograph



The picture of the world we carry shapes what we observe, what we decide and what we do.

Boyd, *The Essence of Winning and Losing*, 1995. Paraphrase

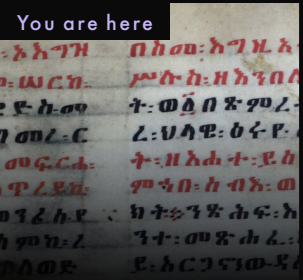
Observation Orientation Decision, action

F-86 Sabres of the 51st Fighter Interceptor Wing over Korea, 1952. U.S. Air Force photograph

# In four blocks

The course, autumn 2026

You are here



## The Word

Large language models

Weekend 1
*16 and 17 October*
Ge'ez games, prompting, OPRO, MLOps

Ethiopian prayer book in Ge'ez, 15th century, Miami University Libraries

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## The Thought

Representation learning

Weekend 2
*13 and 14 November*
Recommendation systems, matrix factorization, bandits

Hortus Deliciarum, c. 1180

## The Power

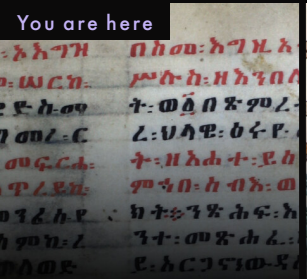
Large-scale computing

Mira (Blue Gene/Q), Argonne National Laboratory, U.S. Department of Energy

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## The Power

Large-scale computing

Mira (Blue Gene/Q), Argonne National Laboratory, U.S. Department of Energy



## The Act

Reinforcement learning

Weekend 3
*11 and 12 December*
MDPs, Bellman equations, SARSA, REINFORCE

F-86 Sabres over Korea, 1952, U.S. Air Force